

Customer Churn Analysis of Luminotech

Dataset of 2023-2024

What We Cover Today

Section 1

Data Preparation — Building churn_df

Section 2

Exploratory Insights — 5 data-driven findings

Section 3

Hypothesis Testing — 3 two-sample t-tests

Section 4

Churn Prediction Model — LR vs Random Forest

Summary

Key findings & management recommendations

01

Data Preparation

Building churn_df

Columns & Construction of churn_df

How churn_df Is Built

Step 1 — Identify customer cohorts

Separate 2023 vs 2024 customers. Exclude new 2024-only customers.

Step 2 — Aggregate per customer

groupby(customer_code).agg(...) on 2023 transactions → one row per customer with spend, frequency, and engagement features.

Step 3 — Derive recency, tenure & margin

recency_days = days from last 2023 purchase to year-end.

tenure_days = last minus first purchase.

avg_margin = profit / spend.

Step 4 — Assign primary district

Most frequent district code per customer mapped to readable name (Sydney, Melbourne, etc.).

Step 5 — Label the target variable

churned = 1 if customer absent from 2024 data, else 0. Missing values imputed via median (numeric) and mode (categorical).

The 16 Columns

Spend & Volume

total_spend · total_profit · avg_txn_value · avg_margin · total_quantity

Engagement & Frequency

n_transactions · n_months_active · n_business_areas · n_distinct_items · n_distinct_salespl

Recency & Tenure

recency_days — days since last 2023 purchase

tenure_days — span from first to last purchase

Geography

district_name — customer's primary sales district (most frequent code, mapped to city name)

Identifiers & Timestamps (EXCLUDED from modelling)

customer_code · last_purchase

Target Variable

churned — 1 = did not return in 2024 (churned), 0 = returned (retained)

churn_df

DATASET OVERVIEW

3,070 customers

12.3%

Churn Rate

★ Key Stat

379 of 3,070 churned

16

Columns

12 numeric, 1 categorical, 1 ID,
1 timestamp + target

50

Average Recency

Ranging from 0–361 days

269

Average Tenure

Engage consistently
throughout the year

Foundation for all subsequent analysis — Sections 2, 3 & 4

02

Exploratory Insights

5 visualisation-based findings from churn_df & the full transaction dataset

Revenue Declined 9.2% — Customer Loss Is the Driver

\$84.9M

2023 Revenue

\$77.1M

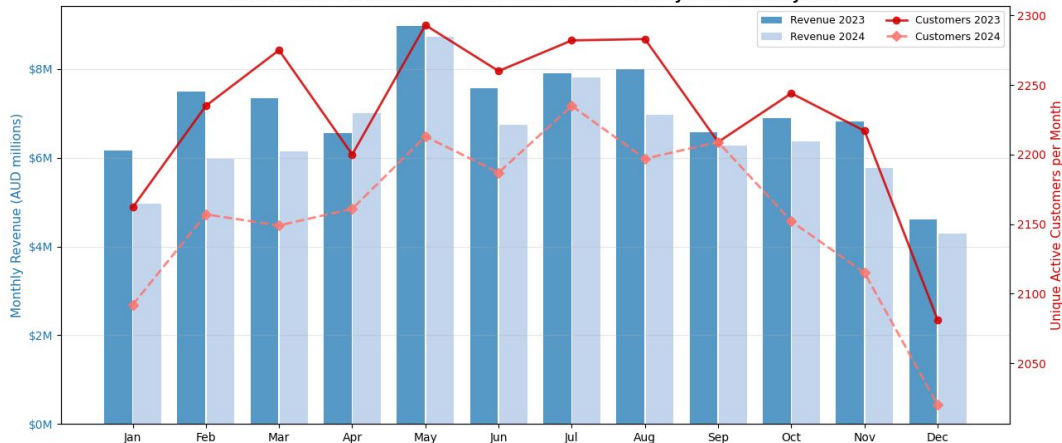
2024 Revenue (−9.2%)

~Perfect

Revenue ↔ Customer correlation

- Monthly revenue tracked customer count almost perfectly across both years
- Decline is driven by losing customers — NOT by existing customers spending less
- May peak visible in both years → seasonal demand cycle exists
- Pre-season promotions could maximise the natural high-demand window

Monthly Revenue and Customer Activity — 2023 vs 2024
Revenue and customer count track each other closely across both years



Recency & Frequency Reveal Two Distinct Customer Clusters

47 days

Recency threshold (P75)

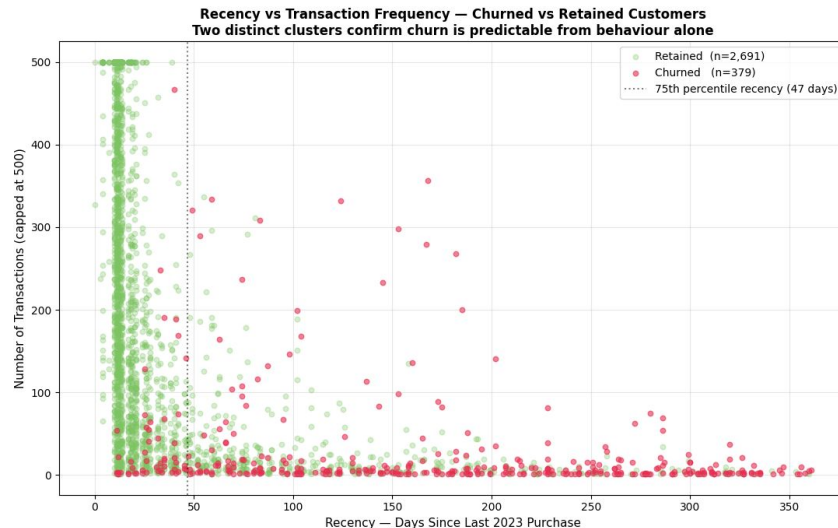
86.3%

Churned beyond threshold

332 vs 28

Avg transactions: retained vs churned

- Retained customers: dense cluster (left side) — low recency, high frequency
- Churned customers: spread right beyond 47-day mark — far fewer transactions
- 86.3% of churned vs only 16.4% of retained exceeded the recency threshold
- Transaction frequency is a strong predictor of churn
- Customer inactivity beyond 47 days serves as an actionable churn-risk signal, enabling earlier intervention and more targeted retention strategies



Churn Is a Behavioural Pattern — Not a Spending Pattern

13 days

Retained median recency

166 days

Churned median recency

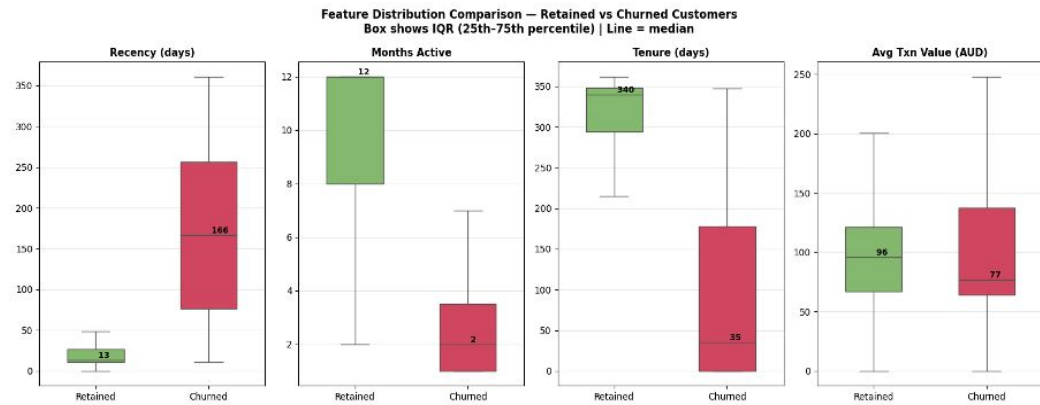
12 months

Retained median active months

2 months

Churned median active months

- Burst-and-exit pattern: churned customers buy in short windows then disappear
- Retained customers maintained purchasing over 340 days vs only 35 for churned
- Avg transaction value gap is modest (\$96 vs \$77) — spending is NOT the signal
- Monitoring recency and months active is far more valuable than monitoring spend



Churn Risk Is Geographically Concentrated

97.5%

Retention in ACT / Riverina

14.5%

Highest Churned Rate

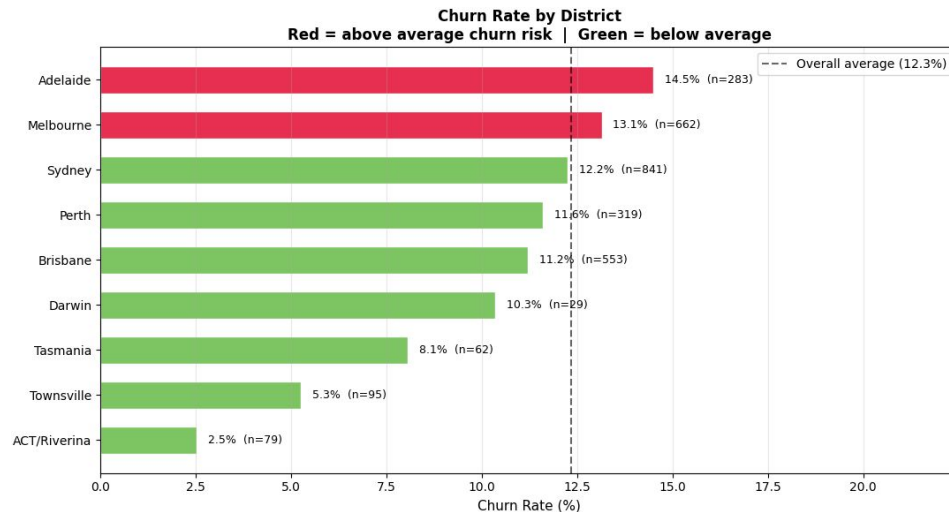
ACT / Riverina

Best Performing District

Adelaide, Melbourne, & Sydney

Drives Most Churn

- Adelaide 14.5% & Melbourne 13.1% are above average
- ACT/Riverina retains 97.5% of customers — best in country
- Sydney & Melbourne = largest volume → most absolute churners
- Investigate ACT/Riverina & Townsville for replicable best practices
- Despite Adelaide's highest rate (14.5%), Sydney loses the most customers in absolute terms (~103), making it the highest-priority intervention target



Feature Correlation: What Actually Predicts Churn?

+0.60

recency_days ↔ churn

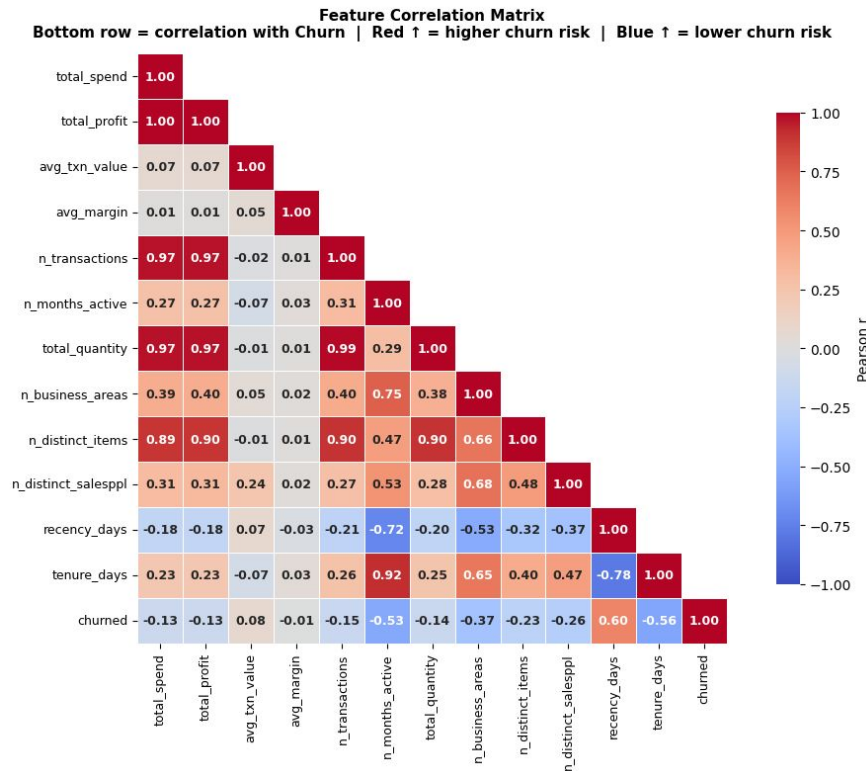
-0.53

n_months_active ↔ churn

-0.56

tenure_days ↔ churn

- Recency is the strongest positive predictor of churn (+0.60)
- Tenure and months active are the strongest negative factors (-0.56, -0.53)
- Some of the volume features are nearly perfectly intercorrelated ($r = 0.89-1.00$)
- Volume features only weakly predict churn (-0.14 to -0.25) — adding them all adds no value
- Implication: monitor engagement frequency and recency, not revenue, as primary loyalty indicators



03

Hypothesis Testing

3 Welch's two-sample t-tests · $\alpha = 0.05$ · All using churn_df

Did Churned Customers Stop Purchasing Earlier?

To determine whether churned customers have a significantly higher number of recency days compared to retained customers.

Null Hypothesis (H0)

Recency days are equal or lower for churned customers ($\mu_{\text{churned}} \leq \mu_{\text{retained}}$)

Alternative Hypothesis (H1)

Churned customers have significantly higher recency days ($\mu_{\text{churned}} > \mu_{\text{retained}}$)

p < 0.001

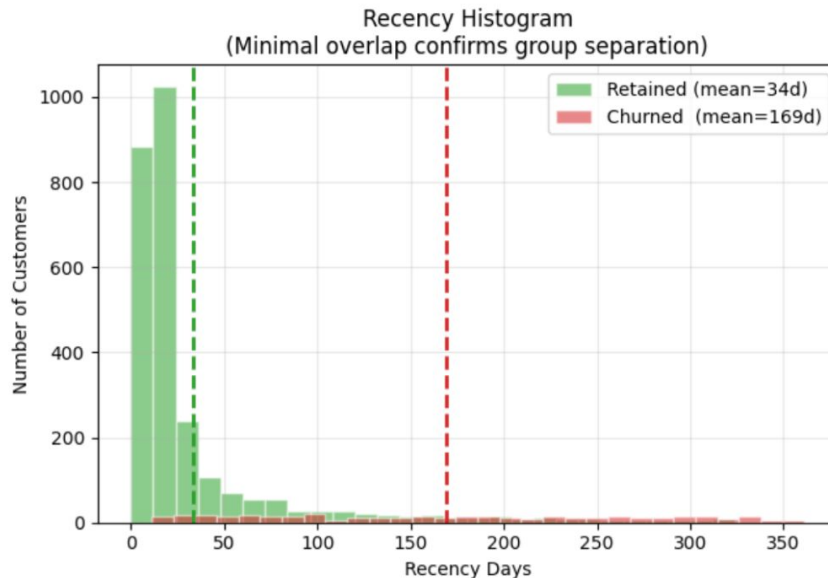
p-value (one-tailed)

✓ **Reject H₀**

p << 0.05

Insights:

- Churned customers: 169 days vs Retained customers: 34 days (**mean gap of 135 days**)
- Customers inactive for 47+ days have an **86.3% probability of not returning (P75 recency)**, a data-derived early warning threshold
- Implementing a **47-day automated inactivity alert** would allow the sales team to proactively intervene before the customer fully disengages



Did Churned Customers Purchase Across Fewer Months?

To determine whether churned customers were active across fewer months than retained customers.

Null Hypothesis (H0)

Churned customers have equal or more active months ($\mu_{\text{churned}} \geq \mu_{\text{retained}}$)

Alternative Hypothesis (H1)

Churned customers have significantly fewer active months ($\mu_{\text{churned}} < \mu_{\text{retained}}$)

$p < 0.05$

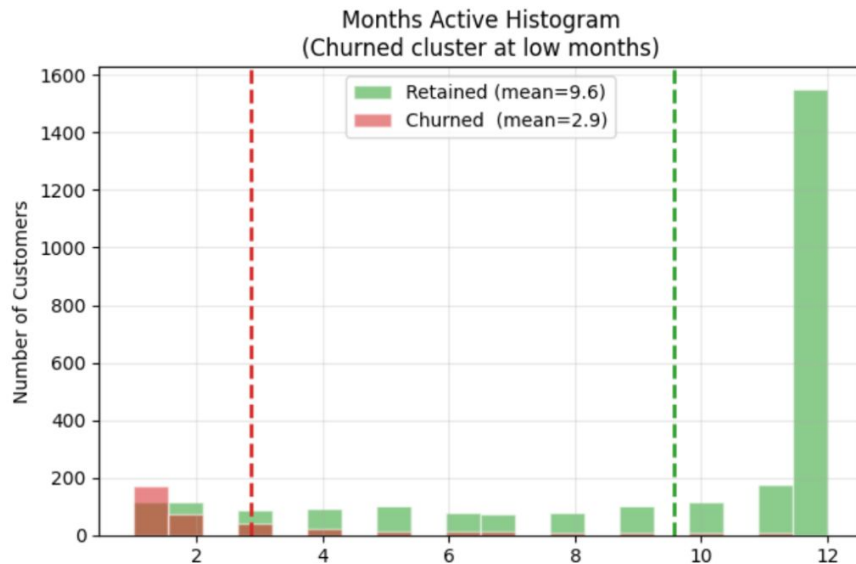
p-value (one-tailed)

✓ **Reject H₀**

$p < 0.05$

Insights:

- **Retained: 9.57 months** vs **Churned: 2.86 months active** (gap of nearly 7 months), confirming a burst-and-exit buying pattern
- Customers who place a second or third order **within the first 2 months (based on churned median)** are significantly more likely to develop a recurring buying habit
- Follow-up outreach after the first order with tailored product recommendations or loyalty incentives can encourage **early repeat purchases**



Is Avg Transaction Value Different Between Groups?

To determine whether average transaction value differs between churned and retained customers.

Null Hypothesis (H0)

No significant difference in average transaction value ($\mu_{\text{churned}} = \mu_{\text{retained}}$)

Alternative Hypothesis (H1)

Significant difference in average transaction value exists ($\mu_{\text{churned}} \neq \mu_{\text{retained}}$)

p < 0.05

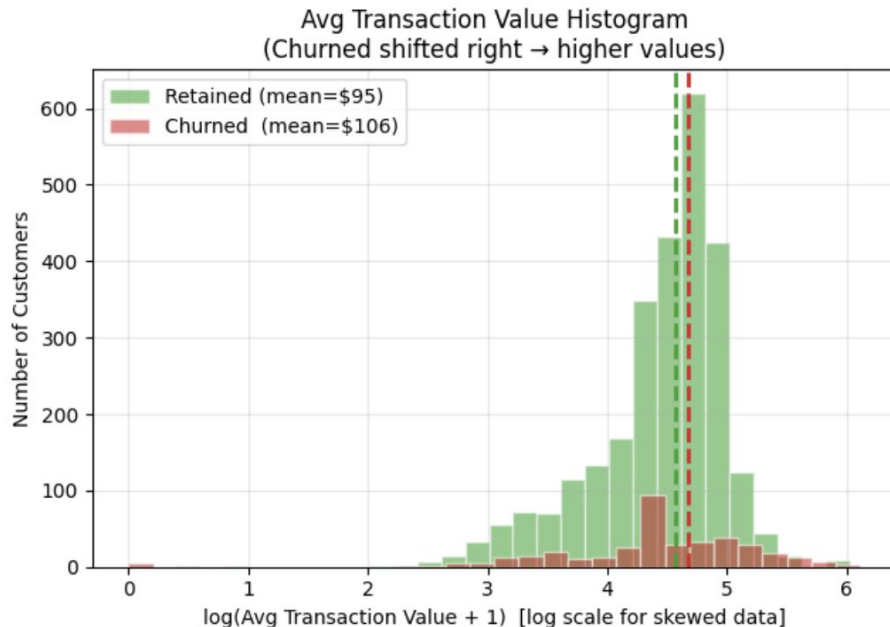
p-value (two-tailed)

✓ **Reject H₀**

Counterintuitive

Insights:

- Churned customers had a **higher average transaction value (\$106 vs \$95)**, proving that high spenders are not necessarily loyal customers
- Churned customers are **project-driven buyers** who place large orders in a short burst before disappearing entirely
- Sales teams should proactively engage large infrequent accounts **during active projects** to secure the next opportunity



04

Churn Prediction

Logistic Regression vs Random Forest · Binary classification · 3,070 customers

What We Are Predicting

Research Question

Which **behavioural features** best predict whether a LuminaTech customer will churn, and how **accurately** can the model identify at-risk customers before they leave?

Target Variable

Churn (binary)

- 1 = did not return in 2024 (churned)
- 0 = returned in 2024 (retained)
- 12.3% churn rate (379 of 3,070 customers)

Key Features

- recency_days
- tenure_days
- n_months_active
- avg_txn_value
- avg_margin
- total_spend
- total_profit
- total_quantity
- n_transactions
- n_business_areas
- n_distinct_items
- n_distinct_salespl
- district_name (one-hot encoded)

Model Performance — Logistic Regression

ROC-AUC = 0.908

The model correctly ranks churned customers above retained 90.8% of the time across all decision thresholds.

77.2%

Recall

★ Primary Metric

Primary — catches 3 in 4 churners

43.3%

Precision

43% of flagged are true churners

55.5%

F1 Score

Balance of Precision & Recall

84.7%

Accuracy

Overall correct predictions

Assumptions Verified

1

No Multicollinearity

✓ SATISFIED

VIF-based sequential removal reduced all features below 5.0. Final set: 15 variables (5 numeric + 10 district dummies).

7 features removed · Highest VIF dropped first each round

total_profit

VIF 920.78

n_transactions

VIF 56.19

n_months_active

VIF 47.47

total_quantity

VIF 28.73

n_distinct_items

VIF 17.82

tenure_days

VIF 15.89

n_distinct_salesppl

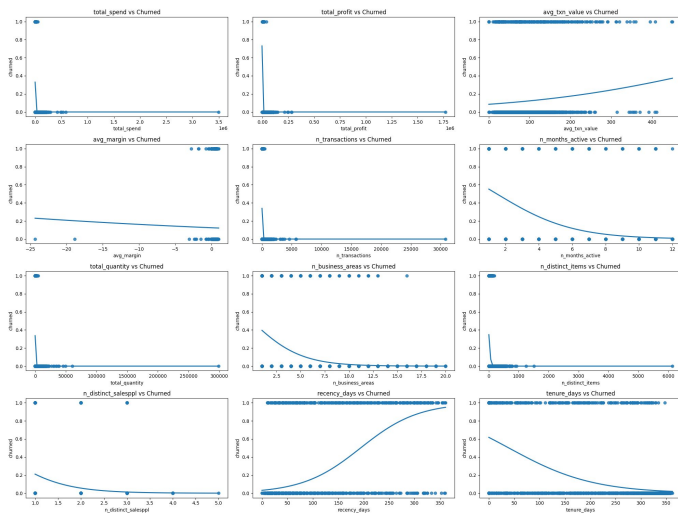
VIF 10.92

Variables VIF

0	total_spend	1.378991
1	avg_txn_value	4.703988
2	n_business_areas	4.863775
3	recency_days	1.958408
4	avg_margin	1.703097
5	district_name_Adelaide	1.622323
6	district_name_Brisbane	2.200567
7	district_name_Darwin	1.081478
8	district_name_Head Office Sales	1.230052
9	district_name_Intercompany Sales	1.138803
10	district_name_Melbourne	2.646981
11	district_name_Perth	1.766937
12	district_name_Sydney	2.896331
13	district_name_Tasmania	1.168256
14	district_name_Townsville	1.245981

Assumptions Verified

recency_days and tenure_days show clean sigmoid curves. Some features appear flat due to right-skewed distributions, rather than violations of the model assumptions.



2

Linearity of the Logit

✓ Reasonably Satisfied

FEATURE ASSESSMENT BY SIGMOID PATTERN

Strong Evidence

recency_days

tenure_days

Moderate Evidence

n_months_active

avg_txn_value

n_business_areas

n_distinct_salespl

avg_margin

Distribution-Limited Assessment

total_spend

total_profit

total_quantity

n_distinct_items

n_transactions

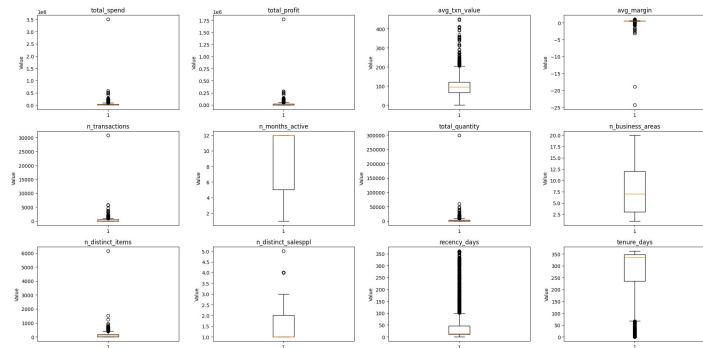
Clear monotonic relationships with churn probability, providing the strongest visual evidence that the assumption is satisfied.

Consistent relationships with churn probability are observed, although some curvature is present. No substantial departures from the assumption are evident.

Relationships appear relatively flat because most observations are concentrated at low values while a small number of extreme observations extend the scale. This pattern reflects strong right-skewness rather than a clear violation of the assumption.

Assumptions Verified

Z-score analysis ($|z| > 3$) identified very low outlier proportions across all features. Although `recency_days` exhibited the highest rate (3.36%; 103 observations), these values represent genuine customer inactivity rather than data errors and were therefore retained.



3

No Extreme Outliers

✓ ACCEPTABLE

OUTLIER SUMMARY (Z-score threshold > 3)

Feature	Outliers %	Interpretation
<code>recency_days</code>	3.36%	Genuine behaviour
<code>avg_txn_value</code>	1.3%	Very low
<code>n_distinct_items</code>	0.72%	Very low
<code>total_quantity</code>	0.39%	Very low
<code>total_spend</code>	0.33%	Very low
<code>n_transactions</code>	0.33%	Very low
<code>avg_margin</code>	0.33%	Very low
<code>total_profit</code>	0.29%	Very low
<code>n_distinct_salespl</code>	0.03%	Very Low
<code>n_business_areas</code>	0.00%	Zero outliers
<code>tenure_days</code>	0.00%	Zero outliers
<code>n_months_active</code>	0.00%	Zero outliers

Model Performance — Random Forest

ROC-AUC = 0.935

The model correctly ranks churned customers above retained 93.5% of the time across all decision thresholds.

37.7%

Recall

Lower Recall vs LR

Catches only 2 in 5 churners

66.2%

Precision

66% of flagged are true churners

48.0%

F1 Score

Balance of Precision & Recall

89.9%

Accuracy

Overall correct predictions

Which Model Should We Deploy?

Logistic Regression		Random Forest	
Accuracy	0.847	0.899 ← better	RF wins
Precision	0.433	0.662 ← better	RF wins
Recall ★	0.772 ← better	0.377	LR wins
F1 Score	0.555 ← better	0.480	LR wins
ROC-AUC	0.908	0.935 ← better	RF wins

Priority Metric

Recall is the business priority — a missed churner costs more than a false alarm.

Recommended Deployment: Logistic Regression

77.2%

Recall — catches 3 in 4 churners

0.908

ROC-AUC — excellent discrimination

88/114

Churners correctly flagged in test

Low cost

False alarm = unnecessary call only

- LR correctly identifies 77.2% of churners vs RF's 37.7% — operationally far more valuable
- RF is valuable for feature importance analysis
- Both models agree: recency and purchase consistency dominate over spend-based features
- Deployment plan: monthly scoring pipeline → accounts above risk threshold → sales team alert
- Model should be retrained quarterly as new transaction data becomes available

What the Data Tells Us — Key Predictors of Churn

Three independent sources of evidence — all pointing to the same top features

Section 2.5

Correlation Heatmap

+0.60 recency
-0.56 tenure
-0.53 months active

Section 3

Hypothesis Tests

$p \approx 0$ for recency
 $p \approx 0$ for months active

Section 4.3

RF Feature Importance

recency 0.220
tenure 0.152
months 0.120

↓ All three approaches converge on the same three behavioural signals ↓

recency_days



0.220

higher value means the customer has not bought in a long time → higher churn risk

tenure_days



0.152

higher value means the customer has been buying over a longer span → lower churn risk

n_months_active



0.120

higher value means the customer purchases across more months → lower churn risk

Key Findings & Recommendations

- 1 Automate a 47-day inactivity alert → immediate sales team outreach
- 2 Focus on habit formation: target repeat purchase within first 60 days
- 3 Do not prioritise by spend — high-value accounts can still churn
- 4 Geo-targeted retention: Adelaide, Melbourne, Sydney need dedicated programmes
- 5 Deploy LR model monthly; retrain quarterly with new data